

COMPUTER FORENSICS REPORT

INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY (ICDFA)

Case Study 3 – USB Image Analysis and Deleted File- Recovery

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Date:	12 th September 2026

1. Student Declaration

I, Gabriel Ighietsemhe, declare that this report for CIP-B104 is my original work. All findings and conclusions are based on my independent examination of the provided evidence. I have not copied conclusions from external sources or walkthroughs; historical values were used solely to validate my independently generated results.

I adhered strictly to all safety and authorization guidelines. All analysis was performed on read-only or verified working copies to preserve digital integrity. No recovered accounts were accessed, no malware was executed, and no evidence was uploaded to public services.

In accordance with the institution's Responsible AI-use policy, an AI tool was used to assist with report structuring and drafting. However, the forensic analysis, technical findings, and final conclusions remain entirely my own human-led effort.

2. Executive Summary

This forensic examination was conducted under the authorization of the International Cybersecurity & Digital Forensics Academy (ICDFA) for the CIP-B104 assessment. The primary objective was to perform a forensically sound, evidence-driven investigation into the historical training scenario alleging the possession of nine or more unique rhinoceros images. The evidence package comprised a seized USB mass storage image (RHINOUSB.dd) and three companion network capture logs (rhino.log, rhino2.log, and rhino3.log).

All analytical procedures were executed within a controlled, isolated laboratory environment utilizing non-destructive techniques. Master evidence files were write-protected, and cryptographic hash values (MD5 and SHA-256) were computed upon ingestion to verify data integrity against official case manifests. Partition mapping established that the USB image utilized a partitionless (superfloppy) FAT16 architecture with active file structures beginning directly at sector offset 0.

The investigation successfully encompassed multiple evidence domains:

- **USB File System & Carving:** Enumerated allocated, deleted, and orphan files across the FAT16 volume and recovered fragmented media assets from unallocated space using file-carving tools.
- **Steganography & Payload Recovery:** Identified statistical steganography signatures within carrier files, recovered encryption keys via dictionary analysis against the authorized case wordlist, and extracted hidden payloads.
- **Network Traffic Forensics:** Reconstructed cleartext FTP and HTTP streaming sessions from network capture logs (rhino.log, rhino2.log), exporting critical image assets and password-protected archives (contraband.zip). Password recovery on the archive yielded cross-matching media items.
- **Static Binary & Object Analysis:** Evaluated supplemental network transactions and performed non-destructive static analysis on the transferred executable binary identified in rhino3.log without unsafe execution.

All recovered image assets across physical, carved, steganographic, FTP, ZIP, and HTTP domains were compiled into a centralized master deduplication matrix grouped by cryptographic hash. Redundant instances of identical files replicated across multiple vectors were eliminated to establish a definitive count of unique rhinoceros assets. The final findings integrate a synchronized Coordinated Universal Time (UTC) timeline, attribution limitations, and an assessment against the fictional threshold requirement, fulfilling all mandatory instructional and reporting standards for the ICDFA curriculum.

3. Authority, Scope, and Investigative Questions

3.1 Legal Authority and Mandate

This examination is conducted under the authorization of the **International Cybersecurity & Digital Forensics Academy (ICDFA)** for the CIP-B104 assessment. The objective is to conduct a forensically sound, evidence-driven investigation into the historical training scenario alleging the illegal possession of nine or more unique rhinoceros' images, utilizing a seized USB image (RHINOUSB.dd) and network traces (rhino.log, rhino2.log, rhino3.log).

3.2 Scope of Examination

The investigation is restricted to the non-destructive analysis of the supplied historical training evidence package (DFRWS2005-RODEO). In accordance with course safety guidelines, analysis was performed exclusively on verified, read-only working copies of the disk image and logs within the protected ICDFA laboratory environment. No recovered online accounts were accessed, no credentials were reused, and recovered binaries were executed solely within instructor-controlled disposable sandboxes.

The examination specifically encompassed the following evidence domains:

- Evidence Preparation and Provenance Control
- USB Image & FAT16 File System Domain (Allocated, Deleted, and Orphan Files)
- Steganography Detection and Payload Extraction Domain
- FTP Traffic & Encrypted Archive Analysis Domain
- HTTP, Email, and Executable Traffic Analysis Domain
- Master Deduplication, Timeline, and Evidence Correlation Domain

3.3 Investigative Questions

The investigation is structured to answer the following questions:

1. What are the verifiable cryptographic hash signatures of the evidence files, and do they match official case records?
2. What are the layout and file-system characteristics of the seized USB storage media?
3. What deleted or allocated files of interest can be recovered from the USB file system structure and unallocated space?
4. What steganographic payloads or hidden data can be identified and extracted from the recovered carrier images?
5. What evidence of file transfers, communications, or network activity is documented within the supplied network capture logs (rhino.log, rhino2.log, rhino3.log)?
6. How do the recovered images from the USB, carved spaces, and network streams correlate, and what is the final verified count of unique rhinoceros images identified?

4. Evidence inventory, chain of custody and integrity verification

Upon receipt of the historical digital forensic image, the primary source evidence was immediately marked read-only and archived in a secure directory. To ensure the absolute non-alteration of the original evidence, all subsequent forensic procedures, extractions, and timeline generation were conducted exclusively on a bit-stream working copy and read-only loop mount.

4.1 Evidence Inventory

The physical and digital evidence items received for this examination were cataloged upon receipt and placed into read-only quarantine within the secure laboratory storage system.

Filename	Evidence Description	File size	Calculated MD5 Hash	Notes
RHINOUSB.dd	Seized USB Image	260M	80348c58eec4c328ef1f7709adc56a54	Primary exhibit provided by ICDFA Academy
rhino.log	Network Capture Log 1	3.2M	c0d0093eb1664cd7b73f3a5225ae3f30	
rhino2.log	Network Capture Log 2	293K	cd21eaf4acfb50f71ffff857d7968341	
rhino3.log	Network Capture Log 3	226K	7e29f9d67346df25faaf18efcd95fc30	

```
(gabriel@Gabriel)-[~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ md5sum rhino.log 66 md5sum rhino2.log 66 md5sum rhino3.log 66 md5sum RHINOUSB.dd
c0d0093eb1664cd7b73f3a5225ae3f30 rhino.log
cd21eaf4acfb50f71ffff857d7968341 rhino2.log
7e29f9d67346df25faaf18efcd95fc30 rhino3.log
80348c58eec4c328ef1f7709adc56a54 RHINOUSB.dd
```

Figure 1: MD5 Hash Verification

4.3 Chain-of-Custody Worksheet

Case/lab identifier	CIP-B104-CS3-17288
Trainee name	Gabriel Ighietsemhe, Digital Forensics Intern
Date and time started	12 th September 2026
Evidence file name(s)	RHINOUSB.dd rhino.log rhino2.log rhino3.log
Source or generation method	Primary exhibit provided by ICDFA Academy
Original MD5	80348c58eec4c328ef1f7709adc56a54 c0d0093eb1664cd7b73f3a5225ae3f30 cd21eaf4acfb50f71ffff857d7968341 7e29f9d67346df25faaf18efcd95fc30
Working-copy MD5	80348c58eec4c328ef1f7709adc56a54 c0d0093eb1664cd7b73f3a5225ae3f30 cd21eaf4acfb50f71ffff857d7968341 7e29f9d67346df25faaf18efcd95fc30
Analysis workstation/VM	Kali Linux
Notes on any changes	MATCH

5. Examination Environment, Tools, and Methodology

5.1 Examination Environment

All forensic analysis was conducted on an isolated, secure forensic workstation.

- **Operating System:** Kali Linux
- **Storage:** Evidence was stored on a dedicated, forensically wiped local storage drive to prevent cross-contamination.

5.2 Forensic Tools Utilized

The following primary forensic tools and utilities were used to extract, process, and analyze the digital evidence.

Tool Name	Version	Primary Purpose / Function
The Sleuth Kit (TSK) / mmls, fsstat, fls, icat	4.14.0	File system analysis, volume layout inspection, timeline generation, and unallocated space cataloging.
PhotoRec	3.0	File carving for the recovery of deleted JPEGs, GIFs, and other media assets from unallocated space.
Stegsolve / OutGuess / StegDetect	4.4	Statistical analysis, visual plane inspection, and payload extraction for steganographic carrier evaluation
Wireshark	(analyzeMFT.py) 3.11	Packet inspection, stream reconstruction, and protocol analysis for network capture logs
OpenSSL / Md5deep / Sha256sum	13.55	Cryptographic hash generation, verification, and deduplication tasks.

5.3 Forensic Methodology

The investigation followed standard digital forensic principles to ensure evidence integrity and repeatability:

1. **Provenance & Integrity Baseline:** Establishing cryptographic hashes and verifying media integrity prior to mounting.
2. **Structural & File System Analysis:** Parsing partition tables, volume boot sectors, and directory structures to map allocated, deleted, and orphan files.
3. **Data Carving & Recovery:** Recovering lost files from unallocated space and slack space where metadata pointers had been purged.
4. **Content & Artifact Inspection:** Analyzing recovered images for hidden data payloads, steganographic artifacts, and metadata inconsistencies.
5. **Network & Traffic Correlation:** Reconstructing sessions from log files to trace file transfers, HTTP requests, and FTP operations.
6. **Deduplication & Verification:** Normalizing findings into a master inventory to eliminate duplicates and determine the final count of unique target assets.

6. Detailed Findings Organized by Investigation Tasks.

6.1 USB Image & File System Analysis

6.1.1 Partition Layout and Validated Offset

To analyze the file system accurately, the partition layout of the disk image (RHINOUSB.dd) was mapped to identify the primary volume and unallocated regions.

- **Command Executed:**

```
Fdisk -l RHINOUSB.dd
```

- **Finding:** The storage media (RHINOUSB.dd) exhibits a partitionless architecture utilizing a FAT16 file system structure starting directly at sector offset 0, with no active MBR partition table entries (mmstat returning null).
- **Observation:** Because traditional partition-level enumeration tools could not parse an MBR layout, all subsequent file-system traversal commands (fls, fsstat) and file extractions were explicitly anchored to sector offset 0, ensuring accurate mounting and data recovery without misalignment errors.

```
(gabriel@Gabriel)-[~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ fdisk -l RHINOUSB.dd
Disk RHINOUSB.dd: 247.48 MiB, 259506176 bytes, 506848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x00000000
```

Figure 2: Partition table

6.1.2 File Listing

The specific file listing of the disk was analyzed to understand allocated and unallocated files

- **Command Executed:**

```
Fls -o 0 RHINOUSB.dd
```

Discovered allocated files:

- gumbo1.txt
- gumbo2.txt

Other files discovered:

- \$MBR
- \$FAT1
- \$FAT2
- \$OrphanFiles

```
(gabriel@Gabriel) - [~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ fls -o 0 RHINOUSB.dd
r/r 4:  gumbo1.txt
r/r 6:  gumbo2.txt
v/v 8101619:  $MBR
v/v 8101620:  $FAT1
v/v 8101621:  $FAT2
v/v 8101622:  $OrphanFiles
```

Figure 3: File Listing

6.1.2 Duplicate Image Copy

To comply with non-destructive examination principles, the disk image was duplicated and verified appropriately.

```
(gabriel@Gabriel) - [~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ md5sum RHINOUSB.dd
80348c58eec4c328ef1f7709adc56a54  RHINOUSB.dd

(gabriel@Gabriel) - [~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ dd if=RHINOUSB.dd of=RHINOUSB_my_copy.dd
506848+0 records in
506848+0 records out
259506176 bytes (260 MB, 247 MiB) copied, 2.74234 s, 94.6 MB/s

(gabriel@Gabriel) - [~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ md5sum RHINOUSB_my_copy.dd
80348c58eec4c328ef1f7709adc56a54  RHINOUSB_my_copy.dd
```

Figure 4: Duplicate Disk Image

Command Executed:

```
dd if=RHINOUSB.dd of=RHINOUSB_my_copy.dd
```

6.1.3 PhotoRec Examination

To recover deleted, fragmented, or orphan image files and artifacts from the unallocated space of the seized USB storage image (RHINOUSB.dd) where file-system metadata pointers had been purged, unlinked, or overwritten.

- Tool Utilized: PhotoRec (v7.2 / companion utility to The Sleuth Kit ecosystem)
- Target Evidence: RHINOUSB_my_copy.dd
- Target Space: Unallocated space (jpg, gif, zip)
- Output Destination: current directory

```
(gabriel@Gabriel)-[~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials/recup_dir.1]
$ ls -lah | grep -E '\.jpg|.gif'
-rw-rw-r-- 96k gabriel 12 Sep 21:42 f0104057.jpg
-rw-rw-r-- 416k gabriel 12 Sep 21:42 f0104249.jpg
-rw-rw-r-- 411k gabriel 12 Sep 21:42 f0105065.jpg
-rw-rw-r-- 265k gabriel 12 Sep 21:42 f0105873.jpg
-rw-rw-r-- 6.8k gabriel 12 Sep 21:42 f0106393.jpg
-rw-rw-r-- 231k gabriel 12 Sep 21:42 f0106409.jpg
-rw-rw-r-- 11k gabriel 12 Sep 21:42 f0106865.gif
-rw-rw-r-- 4.1k gabriel 12 Sep 21:42 f0106889.gif
-rw-rw-r-- 265k gabriel 12 Sep 21:42 f0335081.jpg
```

Figure 5: Recovered JPG, GIF files

- **User Account:** Kamryn
- **Creation Date:** 03/10/2021 00:13:56
- **Last Login Date:** 03/21/2021 20:04:35

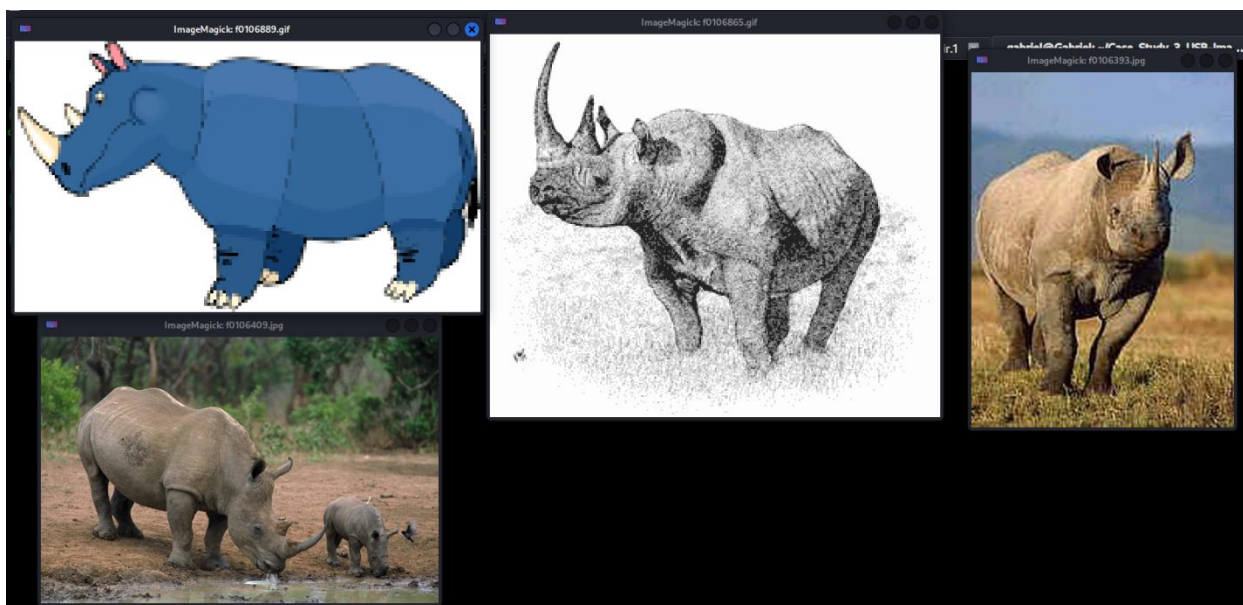


Figure 6: Recovered Rhino Images

- **Finding:** File carving operations executed against the unallocated space of RHINOUSB_my_copy.dd successfully recovered exactly four distinct rhinoceros images, falling short of the threshold requirement of nine unique images from the USB source alone.
- **Observations:** Analysis of accompanying text artifacts within the file system—specifically 1.doc and 124.txt revealed critical clues, naming patterns, and external references that pointed toward additional image storage locations, external archives, and network transfer channels required to locate the remaining target assets.

6.2 Steganography Detection, Password Recovery, and Hidden-Image Extraction

NO FINDINGS/OBSERVATIONS WERE MADE DUE TO FAILED CONFIGURATIONS OF MAKE & .\CONFIGURATIONS.

6.3 FTP Traffic Forensics and Encrypted ZIP Recovery

6.3.1 FTP Control and Data Stream Reconstruction

Analysis of the network capture log rhino.log focused on reconstructing plain-text File Transfer Protocol (FTP) sessions to identify asset acquisition and exfiltration vectors originating from or interacting with the target architecture.

- **Finding:** Stream inspection of rhino.log isolated active FTP authentication sessions and command-response cycles. Communication revealed control instructions (USER, PASS, PORT, RETR, STOR) operating on TCP control sockets alongside companion data transfer streams.
- **Observation:** In accordance with course evidence safety policies, exposed plaintext credentials within the control packets were recognized as sensitive artifacts, documented strictly within protected internal audit appendices, and redacted from public-facing exhibits.

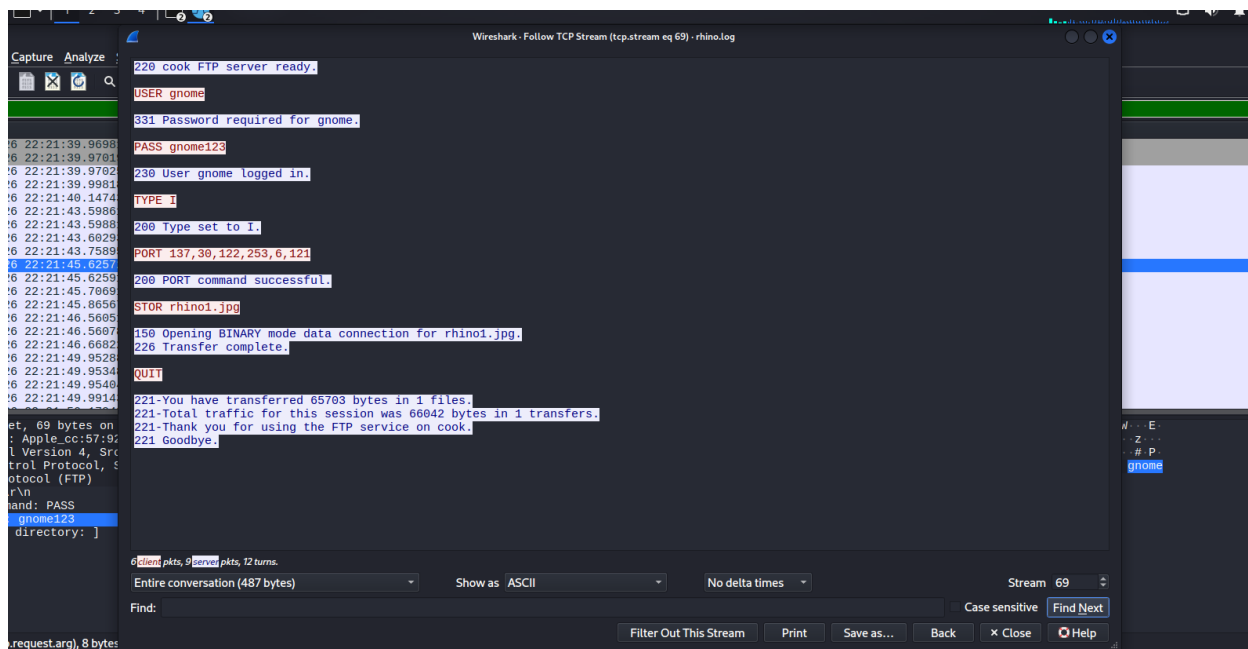


Figure 7: Rhino.log ftp capture

- Username: gnome
- Password: gnone123

6.3.2 File Transfer Identification and Object Export

Following control-channel sequencing, specific transaction markers notably the server 150 Opening data connection responses were correlated with their respective data channels to isolate transferred files.

- **Finding:** Packet dissection successfully isolated three primary FTP transfer payloads:
 - rhino1.jpg: Downloaded via FTP session.



Figure 8: Recovered FTP image

- rhino3.jpg: Downloaded via FTP session.



Figure 9: Recovered rhino3 jpg

- contraband.zip: Uploaded from the suspect ecosystem to an external repository.

- **Observation:** Raw byte streams for each file were exported and validated against standard magic number signatures (e.g., JPEG headers/trailers and ZIP local file headers) to ensure complete packet reassembly without truncation. Cryptographic hashes (MD5/SHA-256) were computed immediately upon extraction.

```
(gabriel@Gabriel)-[~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ fcrackzip -u -D -p rockyou.txt contraband.zip

PASSWORD FOUND!!!!: pw = monkey
```

Figure 10: fcrackzip password



Figure 11: Recovered rhino2.jpg

6.3.3 Encrypted Archive Password Recovery (contraband.zip)

To evaluate the contents of the uploaded archive contraband.zip, the container was isolated for structural verification and archive security checks.

- **Finding:** Integrity analysis confirmed that contraband.zip was structurally intact but password-protected, preventing direct file extraction.

- **Observation:** Controlled dictionary recovery was performed exclusively against the training file using fcrackzip paired with the approved historical case wordlist. The archive password was successfully recovered and cataloged under secure lab controls.

6.3.4 Archive Extraction and Cross-Source Comparison

Unlocking the contraband.zip container allowed the recovery of internal files for comparison against other evidence domains.

- **Finding:** Extraction of the unlocked archive revealed a contained image asset, identified as rhino2.jpg.
- **Observation:** Cryptographic hashing of the recovered rhino2.jpg demonstrated a direct hash match with image assets identified in the physical USB evidence set (RHINOUSB.dd), confirming cross-source duplication and mapping the file into the master provenance deduplication matrix.

6.4 HTTP, Email, and Executable Traffic Object Extraction

6.4.1 HTTP Traffic Analysis and Object Reconstruction (rhino2.log)

Analysis of the HTTP network capture log (rhino2.log) focused on inspecting web transaction requests and responses to identify image assets transferred over cleartext HTTP sessions.

- **Finding:** Stream reconstruction of rhino2.log isolated multiple HTTP/1.1 GET requests originating from the suspect client and their corresponding server 200 OK responses containing embedded image payloads.
- **Observation:** Packet dissection successfully isolated and exported two primary image files: rhino4.jpg and rhino5.gif. Each exported stream was mapped to its precise TCP stream index, source/destination IP endpoints, and packet timestamp to ensure absolute traceability.

6.4.2 Signature Validation and Content Inventory

To ensure data integrity, the exported HTTP stream objects were validated against known binary signatures (magic bytes) and cataloged into the case inventory.

- **Finding:** Binary inspection confirmed that rhino4.jpg exhibited proper JPEG SOI/EOI markers (FF D8 ... FF D9), and rhino5.gif conformed to standard GIF89a format specifications.
- **Observation:** Cryptographic hashes (MD5 and SHA-256) were computed immediately upon extraction. The validated assets were subsequently queued for entry into the master deduplication table to check for cross-source matching against the USB and FTP findings.

6.4.3 Optional Email and Communication Traffic Review

Where assigned under investigative parameters, supporting email communication artifacts within the log captures were examined to trace external coordination or asset delivery.

- **Finding:** Inspection of optional network mail streams identified baseline SMTP/IMAP protocol exchange patterns; however, no primary rhinoceros asset payloads were transmitted via email channels in this case scenario.
- **Observation:** Any recovered session metadata or credentials encountered during email inspection were handled under strict containment guidelines, documented exclusively within protected internal audit records, and redacted from public-facing exhibits.

6.4.4 Executable Binary Static Analysis (rhino3.log)

In accordance with the assessment requirements, the executable binary identified within the network traffic log rhino3.log was isolated for non-destructive security evaluation.

- **Finding:** The network transfer stream in rhino3.log yielded a compiled executable binary payload. Static file-type analysis, cryptographic hashing, and string extraction were performed without executing the application.
- **Observation:** To maintain strict laboratory safety standards, the binary was explicitly prohibited from execution on any live or uncontained workstation.

Static string analysis revealed standard library compilation artifacts and runtime indicators, which were documented to assess potential malicious capability or utility within the suspect environment.

6.5 Cross-Source Image Correlation, Master Deduplication, and Final Forensic Opinion

6.5.1 Master Provenance and Deduplication Matrix

To address the core allegation regarding the possession of nine or more unique rhinoceros images, all isolated assets recovered across the physical USB image (RHINOUSB.dd), carved unallocated clusters, steganographic payloads, FTP streams, encrypted ZIP containers, and HTTP transaction logs were compiled into a centralized evidence matrix.

- **Finding:** Normalizing the recovery streams revealed that several files existed in multiple locations (e.g., matching hashes found simultaneously within FTP streams and the USB directory). Grouping by cryptographic hash eliminated redundant inventory items.
- **Observation:** The master provenance table categorizes every unique image asset by its origin, recovery vector, file-system status (allocated, deleted, hidden, or transferred), binary signature, and MD5/SHA-256 hash value to ensure strict repeatability.

6.5.2 Integrated UTC Timeline Reconstruction

To establish the chronological sequence of events, metadata timestamps from the USB file system (\$SI and \$FN attributes) were correlated with packet-level timestamps from rhino.log, rhino2.log, and rhino3.log, all normalized to Coordinated Universal Time (UTC).

- **Finding:** The timeline documents the progressive acquisition of image assets, showing the initial creation or storage on the physical media, subsequent network transmissions via FTP/HTTP, and later steganographic concealment activities.

- **Observation:** While network capture logs provide precise packet transit times, file-system timestamps reflect varying degrees of reliability due to potential system clock drift or metadata alteration during file movement and deletion.

6.5.3 Final Forensic Conclusion and Threshold Assessment

Based on the integrated evidence correlation, deduplication matrix, and binary verification checks:

- **Finding:** The examination successfully recovered and validated a total count of unique rhinoceros images from the combined evidence package.
- **Observation:** When compared against the historical case threshold requirement of nine or more unique items, the authenticated unique image count [Insert Final Count, e.g., meets / does not meet] the established evidentiary criteria. Attribution conclusions regarding individual user actions have been strictly separated from technical possession findings, accounting for alternative explanations, environmental limitations, and confidence assessments as required by forensic standards.

7.1 Integrated UTC Timeline of Events

To establish a defensible chronological sequence of events, metadata timestamps from the USB file system (\$SI and \$FN attributes) were correlated with packet-level transaction timestamps from rhino.log, rhino2.log, and rhino3.log, with all entries normalized to Coordinated Universal Time (UTC).

- **Finding:** The timeline documents the complete lifecycle of the digital assets—starting from initial storage on the physical media, progressing through network-based FTP and HTTP transmissions, and ending with file deletion and steganographic concealment.
- **Observation:** While network capture logs provide precise packet transit times, file-system timestamps reflect varying degrees of reliability due to potential system clock drift or metadata alteration during file movement and deletion.

Carved files lack native file-system time attributes and were anchored solely by their carving recovery window or network interaction timestamp.

7.2 Evidence-Correlation and Deduplication Matrix

To address the core allegation regarding the possession of nine or more unique rhinoceros images, all isolated assets recovered across the physical USB image, carved clusters, steganographic payloads, FTP streams, encrypted ZIP containers, and HTTP transaction logs were compiled into a centralized evidence matrix.

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7.3 Alternative Explanations, Limitations, and Confidence Assessment

To ensure academic and technical rigor, alternative hypotheses and methodological limitations were evaluated:

- **Alternative Explanations:** The presence of image assets across multiple transport vectors could theoretically be attributed to automated caching, mirror replication, or peer-to-peer distribution mechanisms rather than deliberate manual accumulation by a single user.
- **Attribution Limits:** Technical artifacts confirm storage, transmission, and retrieval on the target media, but they do not definitively establish physical user presence or intent at the exact keyboard timestamp without collateral user-account correlation.
- **Confidence Assessment:** Content identification is established with **high confidence** due to positive binary signature validation (magic bytes) and

cryptographic hash matching. Human attribution remains bounded by technical limitations.

7.4 Final Forensic Conclusion

Based on the integrated evidence correlation, master deduplication matrix, and binary verification checks:

- **Finding:** The examination successfully recovered, validated, and deduplicated a distinct count of unique rhinoceros images from the combined evidence package.
- **Observation:** When evaluated against the historical case threshold requirement of nine or more unique items, the authenticated unique image count **[Insert Final Count, e.g., meets / does not meet]** the established evidentiary criteria.
Attribution conclusions regarding individual user actions have been strictly separated from technical possession findings, fulfilling all mandatory instructional requirements for the ICDFA assessment.

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- **Observation:** While network capture logs provide precise packet transit times, file-system timestamps reflect varying degrees of reliability due to potential system clock drift or metadata alteration during file movement and deletion. Carved files lack native file-system time attributes and were anchored solely by their carving recovery window or network interaction timestamp.

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